THE DUAL-USE HACKATHON · INNOVATELA × USC VITERBI · TRACK 03 RESILIENCE COMMS (RETELL AI)

Turn hold time into context for the next operator

211 Reception is an AI desk for the hold queue of a community and disaster line. While callers wait, Claude talks with them in their language; code, not the model, decides who needs a person first; the operator who picks up gets a summary.

When the fires hit Los Angeles

7,815 contacts offered

60.66% abandoned

1 h 23 min average wait

211 LA, all contacts, 8 Jan 2025 (Eaton and Palisades fires). 211 LA after-action report, App. D.

A first-come queue cannot tell a trapped person from a benefits question. The supervisor asks:

Who needs a person right now?

How it works RUNNING PROTOTYPE

CALLER
Waits in line
Typed chat, or an English phone call via Retell. Never hung up on.

CLAUDE
Talks with them
Editable playbooks: acknowledge, calm, ask one question, in their language.

CLAUDE + CODE
Notes the need
Situation, place, danger, with the caller's words as evidence that code checks.

CODE
Ranks the queue
Fixed rules set the tier, first come within it. On danger the desk gives the emergency number.

PERSON
Takes over
Reserve, then confirm. A failed transfer keeps the original arrival time.

● **Immediate** life threat now ● **Priority** needs a person soon ○ **Unknown** unclear, ranked with Priority ● **Normal** routine, moves up after 30 min

The live demo SCREENSHOTS OF THE RUNNING DESK, SAT 17 OCT 2026

I see smoke coming from a building.

If you or someone with you is in danger right now, call 911 now. Your report has also been flagged for an operator.

Thank you for telling us about the smoke. If anyone may be in danger, please contact 911 now. Where is the building? Please give me the street address or the nearest cross street.

Someone on the second floor is shouting for help and cannot get out.

That sounds frightening, and I am sorry you are dealing with this. Please contact 911 now and tell them someone on the second floor at 12 Oak Street is trapped and calling for help. Is there anyone else inside, and how many people are trapped?

TEAM-WRITTEN SCENARIO **A crisis call.** "I see smoke" makes it Immediate; the desk says call 911 and Claude, live, asks for the address.

call-2 · connecting

Tier: IMMEDIATE, life-threatening now
Waited: 4 min 46 s
Situation: Caller reports smoke coming from a building at 12 Oak Street, and someone on the second floor is shouting for help and cannot get out.
Need: Help for the person on the second floor who cannot get out of the building.
Location: 12 Oak Street
Why this tier: "I see smoke coming from a building."; "Someone on the second floor is shouting for help and cannot get out."
Still unknown: City for the address, Whether there is visible fire or the floor is filling with smoke, Number of people trapped, Whether the trapped person is injured

Confirm human takeover | Transfer failed: return to queue

The handoff. Tier, situation, place, the quotes behind the tier and what is still unknown. Only the operator's confirmation completes it.

call	tier	wait	FIFO position	tiered position	language	origin
call-3	immediate	8m 55s	2	1	en	live
call-5	immediate	4m 39s	4	2	zh	live
call-11	immediate	1m 34s	10	3	es	simulated: team-w
call-19	immediate	1m 15s	18	4	en	simulated: team-w
call-21	immediate	1m 12s	20	5	en	simulated: team-w
call-4	priority	5m 17s	3	6	es	live
call-6	unknown	2m 6s	5	7	en	recorded: Munro d
call-7	priority	1m 53s	6	8	en	recorded: Munro d
call-8	priority	1m 52s	7	9	en	recorded: Munro d
call-9	priority	1m 36s	8	10	en	recorded: Munro d

LABELED SURGE **20 callers in 60 s:** team-written scripts (simulated) and real disaster messages (recorded, not 211 calls). Life threats rise to the top.

Line: 12345 citizen service hotline (China)

我家楼下被水淹了，我奶奶腿脚不方便，出不去，水还在涨

如果您或您身边的人此刻有危险，请立即拨打 110（报警）、119（火警）或 120（急救）。您的情况也已标记给接线员。

听到奶奶被困、水还在涨，我很担心。请你现在立即拨打110、119或120联系紧急电话。请告诉我你们现在的详细地址，比如区、路、小区名和楼号。

金水区花园路88号，3号楼2单元，她在一楼

TEAM-WRITTEN SCENARIO **Twist 2, China 12345 profile.** A flood call in Mandarin: the desk names 110, 119, 120; Claude asks for district, road, compound.

Ordinary day

Callers ask about benefits, rent help, food or road status. The assistant notes the need while they hold, nobody loses their place, and the operator starts with a summary.

Crisis, same build

In a fire or flood surge, life threats move up and hear the emergency number at once. Matching reports are grouped as unverified; silent callers are flagged; dropped callers become callback review tasks.

Twist 2: China's 12345 hotline

	LOS ANGELES 211	CHINA 12345
Languages	English, Spanish, Mandarin	Mandarin, English
In danger	911	110 police · 119 fire and rescue · 120 ambulance
Address	number, street, city	district, road, number, compound, building, unit
Rules	**Identical code:** tiers, order, handoff, human decision	

12345 is China's local government hotline for non-emergency requests; emergencies go to 110, 119, 120 and 122. It surges too: Beijing's 12345 took over 33,000 rain-related calls in 72 hours in July 2023.

Never automated

- Hanging up or calling out
- Dispatch, or promising that help is coming
- Tier, order or handoff decided by the model
- Treating an unclear call as routine
- Completing a transfer without a person

Runs today

- Caller and operator views, saved desk, audit trail
- Claude replies in English, Spanish, Mandarin
- Los Angeles 211 and China 12345 line profiles
- Labeled surge injector
- Retell bridge for English phone calls

Not yet, not measured

- Voice transfer to a person; outbound callback
- Accuracy on real calls; effect on wait times
- Spanish or Chinese phone calls
- A 12345 partner, telephony, hosting, data review

Launch path (Twist 1): DHS S&T SBIR, with AFWERX SBIR Open Topic and NSF SBIR as alternatives. **Built with** FastAPI, Streamlit, SQLite, the Claude API (claude-haiku-5-5) and Retell AI. **AI tools:** Claude Code, OpenAI Codex, ChatGPT.

Data: conversations shown are team-written test scenarios, not real calls. Surge messages: Munro disaster response messages (CC BY 4.0). Arrival timing: Austin APD 911 calls for service. 12345 facts: State Council guidelines 国办发〔2022〕12号 and 国办函〔2025〕66号; Beijing Daily, 2 Aug 2023.